

NAGP AWS CLOUD CASE STUDY

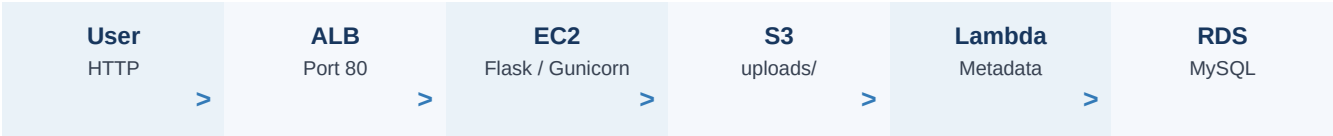
Component Description

Insurance Document Upload MVP | AWS us-east-1 | September 2026

Purpose This document describes the AWS components used to implement a secure, event-driven document upload solution. It explains each component's role, its key configuration, and how the components work together end to end.

1. Solution Overview

The solution accepts document uploads through a public Application Load Balancer while keeping the application compute and database tiers private. Uploaded files are stored in Amazon S3. S3 ObjectCreated events invoke AWS Lambda, which extracts file metadata and persists it to a private Amazon RDS for MySQL database. Secrets Manager provides the database credentials, and CloudWatch provides operational monitoring.



The application tier is distributed across two Availability Zones. The database is intentionally Single-AZ for this training deployment; the DB subnet group spans both Availability Zones so the network design is ready for a Multi-AZ production database.

2. Component Summary

Component	Primary Role	Implementation
Amazon VPC	Network isolation	One VPC (10.20.0.0/16) with six subnets across two Availability Zones.
Application Load Balancer	Public entry point	Internet-facing HTTP listener on port 80; forwards to the private application tier on port 8000.
EC2 + Auto Scaling	Application compute	Two private t3.micro instances at steady state, scaling from 2 to 4 instances.
Amazon S3	Object storage	Private, encrypted storage for application artifacts and uploaded documents.
AWS Lambda	Event processing	Processes S3 upload events and stores document metadata in RDS.
Amazon RDS for MySQL	Metadata database	Private MySQL 8.0 database, db.t3.micro, 20 GiB encrypted storage.
AWS Secrets Manager	Credential storage	RDS-managed database credentials; password is not embedded in code or Lambda environment variables.
VPC Endpoints	Private AWS service access	S3 Gateway Endpoint plus Secrets Manager and CloudWatch Logs Interface Endpoints.
IAM	Least-privilege authorization	Separate human and workload permissions for the app, Lambda, and build process.
Amazon CloudWatch	Monitoring and alerting	Central dashboard and alarms for ALB, ASG, Lambda, and RDS health.
AWS CloudFormation	Infrastructure as Code	Reproducible foundation and application stacks with explicit lifecycle controls.

3. Network and Connectivity Components

Network Component	Description
VPC	10.20.0.0/16 in us-east-1. Provides the isolation boundary for all networking resources.
Public subnets	Two /24 subnets, one in each Availability Zone. Host the internet-facing ALB network interfaces.
Private application subnets	Two /24 subnets, one per Availability Zone. Host EC2 application instances and Lambda VPC networking.
Private database subnets	Two /24 subnets, one per Availability Zone. Used by the RDS DB subnet group.
Internet Gateway	Attached to the VPC. Only the public route table has a 0.0.0.0/0 route to the Internet Gateway.

Network Component	Description
NAT Gateway	Not deployed to avoid unnecessary training-environment cost. Private AWS service access is provided through VPC endpoints instead.
S3 Gateway Endpoint	Provides private S3 connectivity without an internet route or NAT Gateway.
Secrets Manager Interface Endpoint	Provides private access to the database credential service through endpoint ENIs in both Availability Zones.
CloudWatch Logs Interface Endpoint	Allows private workloads to send logs without requiring public internet access.
Network ACLs	Separate public, private-application, and private-database NACL tiers. The application tier includes required ephemeral TCP/UDP handling for VPC-attached Lambda networking.

4. Application and Compute Components

Compute Component	Description
Application Load Balancer	Internet-facing ALB spanning both public subnets. Accepts HTTP on port 80 and forwards requests to the application target group.
Target Group	Routes traffic to EC2 on port 8000. The /health endpoint is used for health checks.
EC2 application instances	Private t3.micro Linux instances with no public IP address and no SSH key pair. The application runtime uses Python 3.11, Flask, and Gunicorn.
Custom AMI	Prebuilt application image used by the Auto Scaling launch template for consistent instance startup.
Auto Scaling Group	Minimum 2, desired 2, maximum 4. Instances are distributed across both private application subnets.
Scaling policy	Target tracking based on approximately 50% average CPU utilization.
Instance Metadata Service	IMDSv2 is required on application instances to reduce metadata-service exposure.
Application behavior	GET / serves the upload UI, GET /health provides the ALB health check, and POST /upload writes files to the S3 uploads/ prefix.

5. Storage and Event Processing Components

Storage / Event Component	Description
Amazon S3 bucket	Private bucket with Block Public Access enabled, BucketOwnerEnforced object ownership, TLS-only access policy, and SSE-S3 default encryption.
artifacts/app/	Stores the packaged EC2 application artifact used during image preparation.

Storage / Event Component	Description
artifacts/lambda/	Stores the packaged Lambda deployment artifact.
uploads/	Stores user-uploaded documents. Only this prefix participates in the business event-processing flow.
S3 event notification	ObjectCreated events for uploads/ invoke the document-processing Lambda. Artifact uploads do not invoke the function.
Lambda function	Python 3.14 function. Reads S3 object metadata, including filename, content type, upload time, bucket, and object key.
Database write	Lambda uses PyMySQL to create/insert the metadata record. Processing is idempotent for the same bucket, key, and upload timestamp.
Controlled read mode	A read_recent action can retrieve recently persisted metadata records for demonstration and validation without exposing database credentials.

6. Database and Secrets Components

Database / Secret Component	Description
Amazon RDS for MySQL	MySQL 8.0 database using db.t3.micro and 20 GiB gp2 storage. Encryption at rest is enabled.
Private placement	RDS is not publicly accessible and is deployed into the private DB subnet group.
Single-AZ deployment	Used deliberately for the training/free-plan environment. This document does not claim database high availability.
DB subnet group	Spans both private database subnets, preserving a clean path to a future Multi-AZ production design.
Secrets Manager	RDS manages the master credentials in Secrets Manager. The password is not stored in source code, CloudFormation parameters, or Lambda environment variables.
Lambda database access	Only the Lambda security group is permitted to connect to the RDS security group on MySQL port 3306.
Application database access	The EC2 application tier has no direct MySQL security-group path to RDS.

7. Security and Access-Control Components

Security Component	Description
ALB security group	Inbound HTTP port 80 from the internet; egress to the application tier on port 8000.
Application security group	Inbound port 8000 only from the ALB security group. No inbound SSH rule.

Security Component	Description
Lambda security group	No inbound rule. Outbound access is limited to the connectivity needed for RDS and private AWS service endpoints.
RDS security group	Inbound MySQL port 3306 only from the Lambda security group.
Human IAM group/user	A project developer group and user demonstrate restricted project permissions rather than administrator access.
EC2 application role	Provides only the permissions the web application needs, primarily PutObject access to uploads/.
Lambda execution role	Provides S3 read access for uploads/, permission to read the specific database secret, CloudWatch logging, and VPC ENI operations.
Builder role	Limited to reading the application artifact required to build the custom AMI.
Credential handling	No long-lived application credentials are embedded in source code or deployment artifacts. Workloads use IAM roles.
S3 controls	Public access is blocked, encryption is enabled by default, ownership is bucket-enforced, and insecure transport is denied.

8. Monitoring and Operations Components

Operations Component	Description
CloudWatch dashboard	The nagp-overview dashboard brings together operational metrics from ALB, Auto Scaling, Lambda, and RDS.
Lambda error alarm	Detects Lambda processing errors.
ALB 5XX alarm	Detects application-facing server-side errors at the load balancer.
Unhealthy target alarm	Detects unhealthy ALB target-group members.
ASG in-service-low alarm	Detects application capacity falling below the intended in-service level.
RDS CPU alarm	Detects sustained high database CPU utilization.
RDS connections alarm	Detects unusually high database connection count.
CloudFormation	Foundation and application templates define the infrastructure consistently and support repeatable creation and teardown.

9. End-to-End Processing Flow

Step	Description
1. Client request	A user opens the application through the public ALB over HTTP.

Step	Description
2. Application routing	The ALB routes the request to a healthy private EC2 instance on port 8000.
3. Upload	The Flask application writes the file to the private S3 bucket under uploads/<unique-id>/<filename>.
4. Event	S3 emits an ObjectCreated notification only for the uploads/ prefix.
5. Processing	Lambda receives the event and reads the object metadata.
6. Credential retrieval	Lambda retrieves the RDS-managed database credentials through the private Secrets Manager endpoint.
7. Database write	Lambda connects to private RDS on port 3306 and inserts the document metadata record.
8. Observability	CloudWatch captures metrics, logs, and alarm state for the major runtime components.

10. Design Choices and Training Constraints

The implementation intentionally separates production-oriented security patterns from cost-sensitive training decisions. These choices are part of the design and are not hidden limitations.

Decision / Constraint	Rationale and Production Direction
HTTP instead of HTTPS	The ALB exposes HTTP port 80 for the assignment demonstration. A production deployment should use HTTPS with an ACM certificate and redirect HTTP to HTTPS.
No NAT Gateway	Avoids ongoing NAT cost. Required AWS service access is handled with S3, Secrets Manager, and CloudWatch Logs VPC endpoints.
Single-AZ RDS	Meets the training scope and cost constraints. A production implementation should use Multi-AZ RDS for database high availability.
No end-user authentication	The MVP focuses on cloud/network design and document-event processing. Production should add authentication and authorization.
No WAF or malware scanning	Recommended production hardening would add AWS WAF and a file-security scanning workflow before downstream use of uploaded documents.
Application tier redundancy	Unlike the database tier, the application tier is deliberately spread across two Availability Zones and managed by Auto Scaling.

11. Component Responsibility Matrix

Responsibility	AWS Component
Public traffic entry	Application Load Balancer
Application execution	EC2 Auto Scaling Group

Responsibility	AWS Component
Uploaded file storage	Amazon S3
Upload event generation	Amazon S3 Event Notifications
Metadata extraction	AWS Lambda
Credential storage	AWS Secrets Manager
Metadata persistence	Amazon RDS for MySQL
Private AWS-service connectivity	VPC Endpoints
Network isolation	VPC, subnets, route tables, NACLs, security groups
Identity and authorization	AWS IAM
Metrics, logs and alerts	Amazon CloudWatch
Provisioning and teardown	AWS CloudFormation

12. Summary

The architecture combines a public load-balancing layer with private compute, event-driven serverless processing, private database access, managed secrets, least-privilege IAM, private service endpoints, and centralized monitoring. The separation of responsibilities ensures that the EC2 web tier handles uploads, Lambda handles metadata processing, and only Lambda can reach the database. This keeps the MVP simple while demonstrating the required cloud computing, networking, security, automation, and observability concepts.

Key design statement: The public ALB is the only internet-facing application component. EC2, Lambda networking, and RDS remain private, and the EC2 application has no direct database access.